Project 2 Report

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Contributions

Liam Stevens: 33%

Cassini Rajesh: 33%

Adam Skutnik: 33%

Honor Statements

I, Liam Stevens, have neither given nor received any assistance on this project.

I, Cassini Rajesh, have neither given nor received any assistance on this project.

I, Adam Skutnik, have neither given nor received any assistance on this project.

Introduction

Simulating how an aircraft moves provides insight as to how it will perform and is vital to designing and improving it. For this project, a quadcopter will be simulated which consists of four equally spaced rotors and has six degrees of freedom. Two frames will be used to represent the quadcopter's position in space, an inertial and body frame. The body frame is attached to the quadcopter and the inertial frame is fixed to the ground with gravity acting in the negative-z direction. A schematic of the quadcopter is provided below in Figure 1.

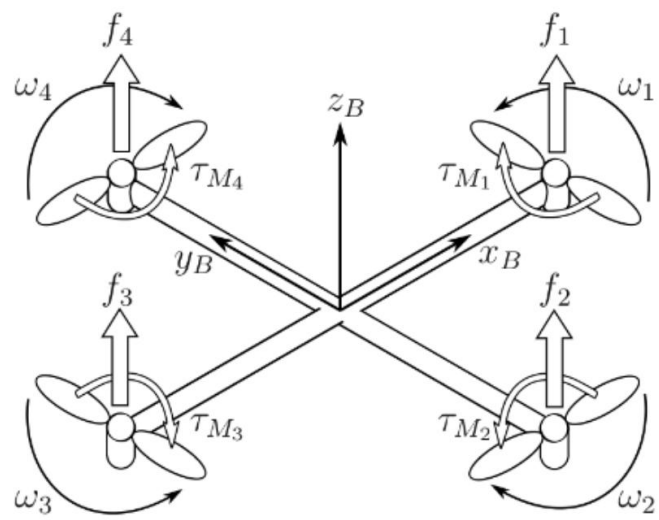

Figure 1. *Schematic of the modelled quadcopter.*

In Figure 1, the four motors are all spaced out equally from the center of the quadcopter with length l. The body frame basis is shown in Figure 1 which is fixed in the center of each support. At the ends of each support, the rotor rates, reaction moment, and thrust per rotor is shown.

Constants

Constants representing the properties of the drone and the gravitational constant is shown below.

$g = 9.81\ \frac{m}{s^2}$

$m = 0.450\ kg$

$l = 0.225m$

$k = 2.98\ \times\ 10^{-7}$

$b = 1.14\ \times\ 10^{-6}$

$I_{xx} = I_{yy} = 4.85\ \times 10^{-3}\ kg\ m^2$

$I_{zz} = 8.80\ \times 10^{-3}\ kg\ m^2$

Simulating Quadcopter Motion

The full maneuverability of the quadcopter will take six seconds as it goes through different movements. To begin, the quadcopter will ascend for the first second of the simulation by increasing all the rotor velocities equally. Individual angular velocities of the rotors are provided below.

$$\Omega_i = \Omega_{i,hover} + 70\sin\left(\frac{2\pi t}{4}\right), \qquad 0 \le t < 1$$

$$\Omega_{i,hover} = \sqrt{\frac{mg}{4k}}$$

For the following second, the ascension is stopped by decreasing the rotor velocities equally.

$$\Omega_i = \Omega_{i,hover} - 77\sin\ \ [\![(2\pi t/4)]\!]\ , \qquad 1 \le t \le 2$$

The third second consists of putting the quadcopter into a roll motion which is achieved by increasing the velocity of the fourth motor while simultaneously decreasing the velocity of the second motor.

$$\Omega_2^2 = \Omega_{2_{hover}}^2 - 70^2\sin\left(\frac{2\pi(t-2)}{4}\right), \qquad 2 < t < 3$$

$$\Omega_4^2 = \Omega_{4_{hover}}^2 + 70^2 \sin\left(\frac{2\pi(t-2)}{4}\right), \qquad 2 < t < 3$$

Then the roll rate is reduced back to zero over the course of the fourth second by doing the opposite maneuver.

$$\Omega_2^2 = \Omega_{2_{hover}}^2 + 70^2 \sin\left(\frac{2\pi(t-2)}{4}\right), \qquad 3 \leq t < 4$$

$$\Omega_4^2 = \Omega_{4_{hover}}^2 - 70^2 \sin\left(\frac{2\pi(t-2)}{4}\right), \qquad 3 \leq t < 4$$

Once the fifth second begins, the quadcopter enters a pitch motion by increasing the velocity of the third rotor and decreasing the velocity of the first rotor.

$$\Omega_1^2 = \Omega_{1_{hover}}^2 - 70^2 \sin\left(\frac{2\pi(t-4)}{4}\right), \qquad 4 \leq t < 5$$

$$\Omega_3^2 = \Omega_{3_{hover}}^2 + 70^2 \sin\left(\frac{2\pi(t-4)}{4}\right), \qquad 4 \leq t < 5$$

Finally, the pitch motion is reversed by doing the opposite maneuver over the sixth second.

$$\Omega_1^2 = \Omega_{1_{hover}}^2 + 70^2 \sin\left(\frac{2\pi(t-4)}{4}\right), \qquad 5 \leq t \leq 6$$

$$\Omega_3^2 = \Omega_{3_{hover}}^2 - 70^2 \sin\left(\frac{2\pi(t-4)}{4}\right), \qquad 5 \leq t \leq 6$$

The full motion of the quadcopter over these six seconds is provided below in Figures 2 through 5.

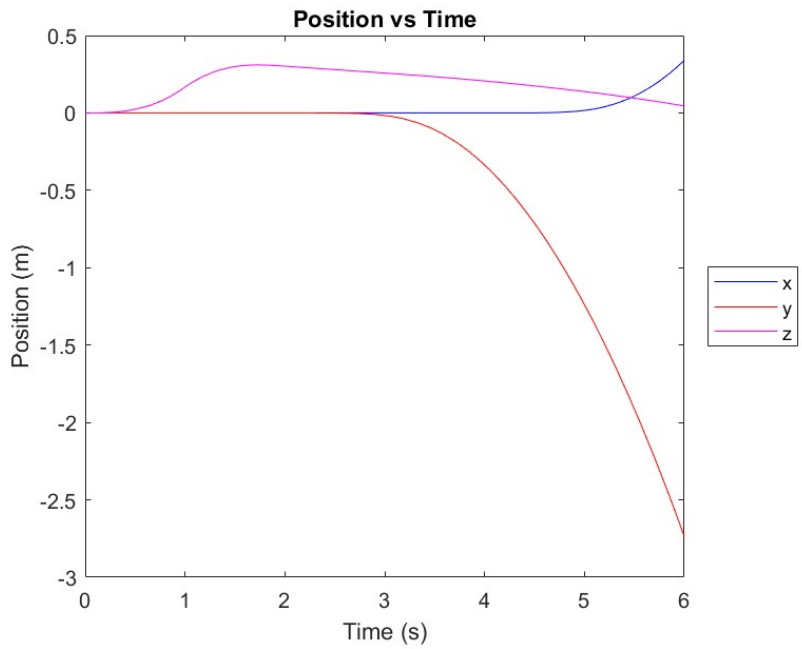

Figure 2. *Position of the modelled quadcopter in inertial frame.*

As seen in Figure 2, the quadcopter reaches a hover altitude of 0.3 meters. After the quadcopter goes into its roll motion, its altitude starts to decrease at a slow rate, and it begins to move in the negative-y direction at a rapid rate. Then, afterwards, the quadcopter goes into its pitch motion, it begins to move in the positive x direction. However, it's movement in the x direction is at a much lower rate than the y direction.

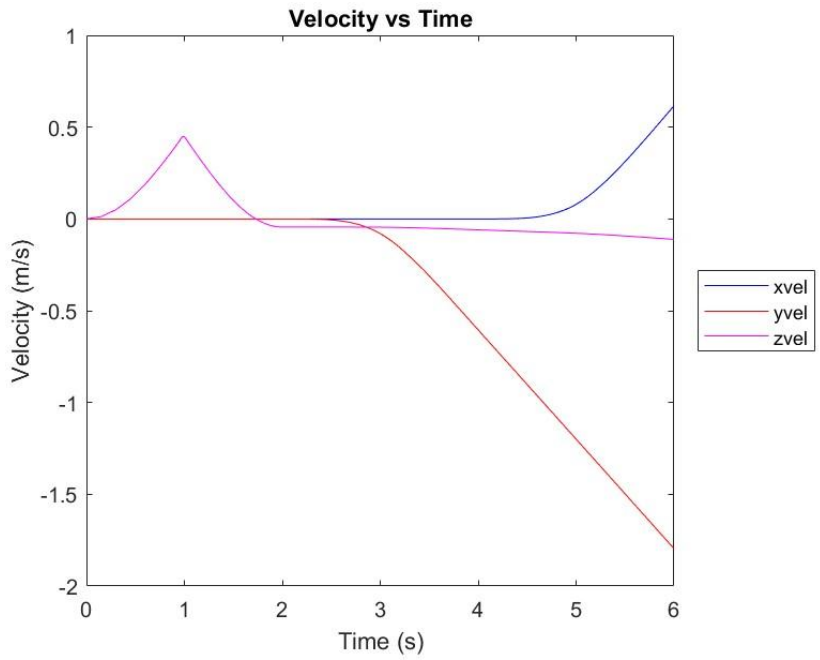

Figure 3. *Velocity of the modelled quadcopter in body frame.*

Figure 3 depicts the velocity of the quadcopter as it goes through its maneuvers. As expected, its z velocity increases greatly while it's ascending and then decreases again as it approaches its hovering point. Then, at the third second, the quadcopter moves in the negative-y direction as it goes through a rolling motion. Finally, after 5 seconds, the quadcopter moves in the positive-x direction after pitching.

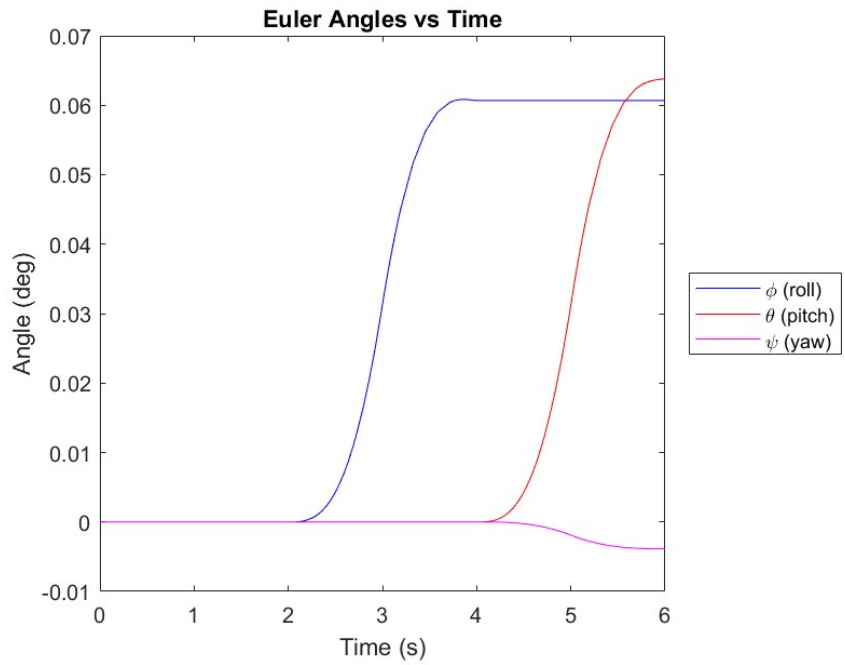

Figure 4. *Euler angles of the modelled quadcopter.*

As seen in Figure 4, the drone initially rolls up then levels out from the initial. Then after levelling, the drone pitches up from the initial and yaws right from initial.

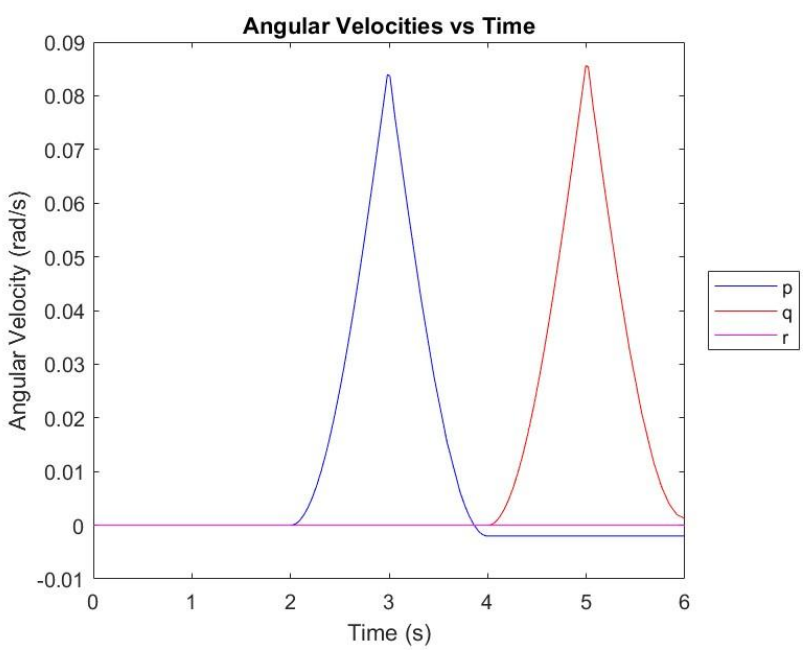

Figure 5. *Angular velocity of the modelled quadcopter.*

Figure 5 shows the angular velocities of the modelled quadcopter as it goes through the maneuvers. As expected, there are no angular velocities when the quadcopter is ascending. Then, once it goes through a roll, p increases at first and then decreases as it stops the roll. This behavior is seen again in the q once the quadcopter goes through the pitch.

Quadcopter Hovering Autopilot

Consider the following equations for an autopilot:

$$\mathrm{T} = \left(\mathrm{g} + (\dot{\mathrm{z}}_{\mathrm{r}} - \dot{\mathrm{z}}) + (\mathrm{z}_{\mathrm{r}} - \mathrm{z})\right)\frac{\mathrm{m}}{\cos\phi\cos\theta}$$
$$L = I_{xx}\left(\left(\dot{\phi}_r - \dot{\phi}\right) + (\phi_r - \phi)\right)$$
$$M = I_{yy}\left(\left(\dot{\theta}_r - \dot{\theta}\right) + (\theta_r - \theta)\right)$$
$$N = I_{zz}\left(\left(\dot{\psi}_r - \dot{\psi}\right) + (\psi_r - \psi)\right)$$

Where the subscript 'r' represents a reference signal. The angular velocities of the rotor are computed with the autopilot as follows:

$$\Omega_1^2 = \frac{T}{4k} - \frac{M}{2kl} + \frac{N}{4b}$$
$$\Omega_2^2 = \frac{T}{4k} - \frac{L}{2kl} - \frac{N}{4b}$$
$$\Omega_3^2 = \frac{T}{4k} + \frac{M}{2kl} + \frac{N}{4b}$$
$$\Omega_4^2 = \frac{T}{4k} + \frac{L}{2kl} - \frac{N}{4b}.$$

The derivation for each angular velocity is as follows. We know that the following relationships from the equations of motion are true:

$$T = k\sum_{i=1}^{4}\Omega_i^2$$

(i)

$$L = kl(-\Omega_2^2 + \Omega_4^2)$$

$$M = kl(-\Omega_1^2 + \Omega_3^2) \tag{ii}$$

$$N = b(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2) \tag{iii}$$

(iv)

First, we will derive the odd numbered index of Ω_i^2. We divide equation (i) by k and then add $\frac{N}{b}$ which will result in the following equation.

$$\frac{T}{k} + \frac{N}{b} = 2\Omega_1^2 + 2\Omega_3^2$$

$$\frac{T}{2k} + \frac{N}{2b} = \Omega_1^2 + \Omega_3^2.$$

Then, $\frac{M}{kl}$ is subtracted from the equation as seen below.

$$\frac{T}{2k} + \frac{N}{2b} - \frac{M}{kl} = 2\Omega_1^2 \Rightarrow \Omega_1^2 = \frac{T}{4k} + \frac{N}{4b} - \frac{M}{2kl} \ .$$

To find Ω_3, $\frac{M}{kl}$ is added which results in the following.

$$\frac{T}{2k} + \frac{N}{2b} + \frac{M}{kl} = 2\Omega_3^2 \Rightarrow \Omega_3^2 = \frac{T}{4k} + \frac{N}{4b} + \frac{M}{2kl} \ .$$

Even indexed Ω_i are derived as follows. If we subtract equation (i) over k by $\frac{N}{b}$ it will result in the equation below.

$$\frac{T}{2k} - \frac{N}{2b} = \Omega_2^2 + \Omega_4^2.$$

Subtracting the equation by $\frac{L}{kl}$ results in the following.

$$\frac{T}{2k} - \frac{L}{kl} - \frac{N}{2b} = 2\Omega_2^2 \Rightarrow \Omega_2^2 = \frac{T}{4k} - \frac{L}{2kl} - \frac{N}{4b}.$$

If we added $\frac{L}{kl}$ to the equation it results in the following as well.

$$\frac{T}{2k} + \frac{L}{kl} - \frac{N}{2b} = 2\Omega_4^2 \Rightarrow \Omega_4^2 = \frac{T}{4k} + \frac{L}{2kl} - \frac{N}{4b}.$$

Furthermore, the angular rates of each rotor can be computed as shown below.

$$\begin{aligned}
\Omega_1^2 &= \frac{T}{4k} - \frac{M}{2kl} + \frac{N}{4b} \\
\Omega_2^2 &= \frac{T}{4k} - \frac{L}{2kl} - \frac{N}{4b} \\
\Omega_3^2 &= \frac{T}{4k} + \frac{M}{2kl} + \frac{N}{4b} \\
\Omega_4^2 &= \frac{T}{4k} + \frac{L}{2kl} - \frac{N}{4b}.
\end{aligned}$$

With all the rotor angular velocities found, a simulation for the quadcopter's auto pilot can be run. In this simulation, the autopilot is to reach and maintain a height (z-axis) of 10 meters over the course of 120 seconds. The initial conditions of the simulation are provided below, and all other reference signals are equal to 0.

$$x = y = \dot{x} = \dot{y} = \dot{z} = p = q = r = 0$$

$$z = 1m, \qquad z_r = 10m$$

$$\theta = \phi = \psi = 10^o$$

The results of the simulation are provided below in Figures 6 through 10.

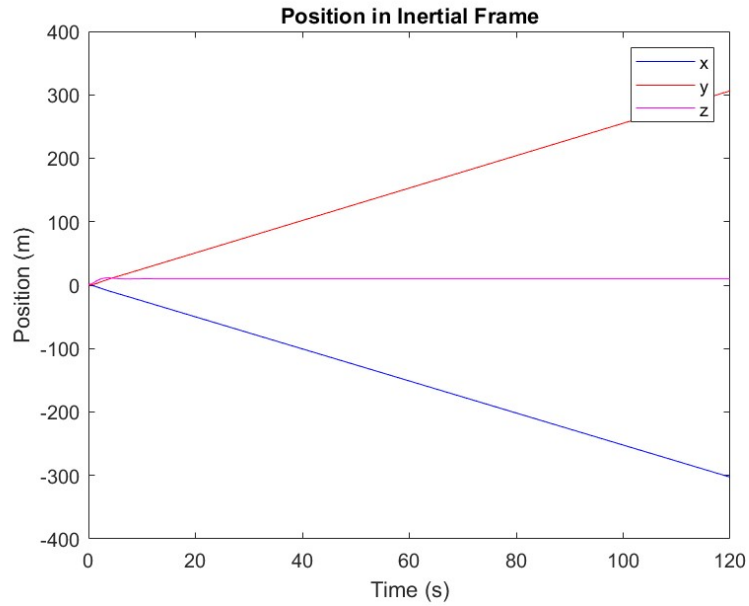

Figure 6. *Position of the modelled quadcopter in inertial frame.*

As seen in Figure 6, the quadcopter remains at a constant altitude after reaching the desired height. It is also noted that the quadcopter moves equally in the x and y direction while staying hovered.

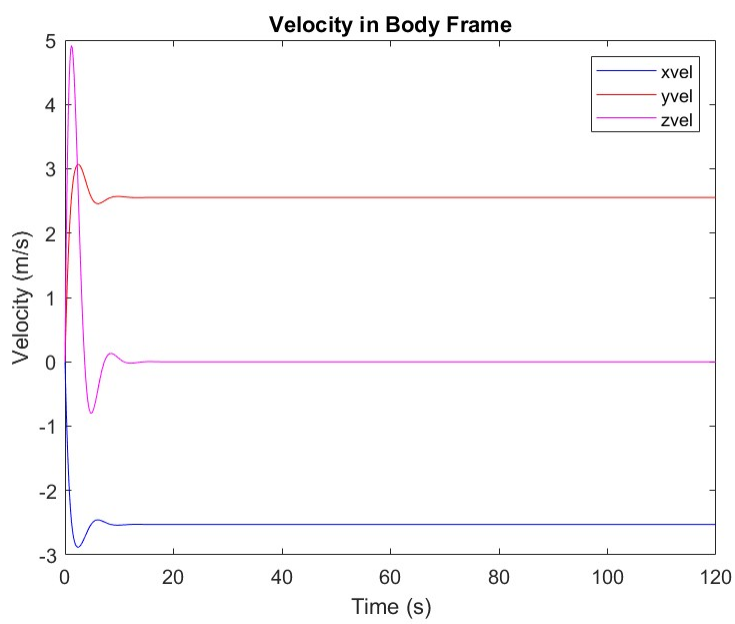

Figure 7. *Velocity of the modelled quadcopter in body frame.*

In Figure 7, each desired velocity in the autopilot follows the same overshoot, undershoot, and then asymptote at the desired velocity.

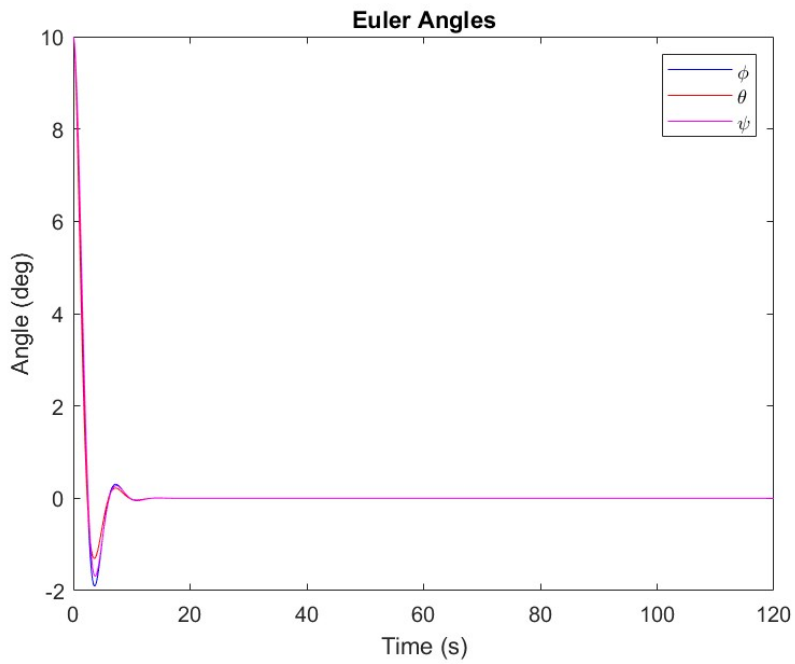

Figure 8. *Euler angles of the modelled quadcopter.*

In Figure 8, the curve overshoots, undershoots and then finally asymptotes at the desired angle.

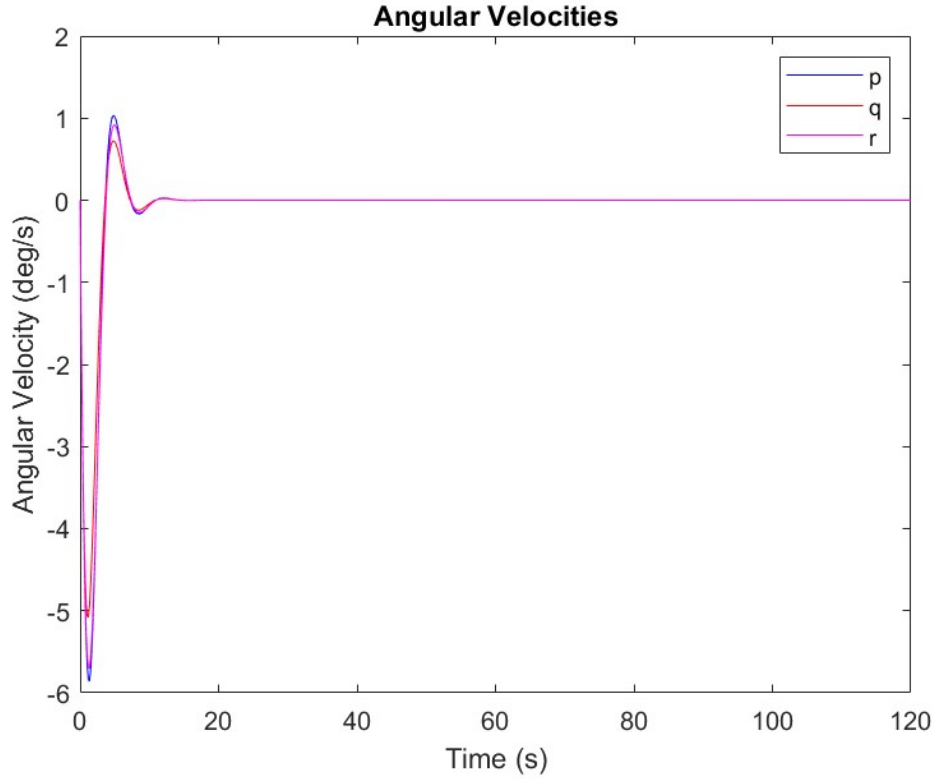

Figure 9. *Angular velocity of the modelled quadcopter.*

We can see, in Figure 9, the angular velocity overshoots, undershoots and then asymptotes to the reference point.

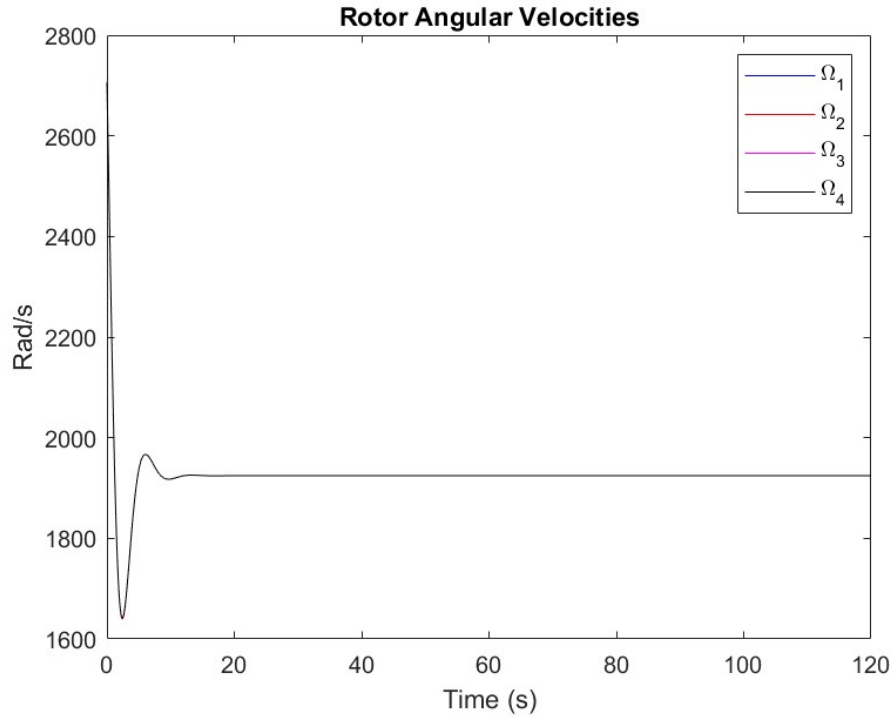

Figure 10. *Rotor angular velocity of the modelled quadcopter.*

Figure 10 depicts the rotor angular velocity as the quadcopter hovers. Similarly to Figure 8 and 9, the rotor angular velocity undershoots, overshoots, and finally settles at a reference value.

For each autopilot curve, the dependent variables follow an overshoot, undershoot, and then an asymptotic behavior as it approaches the reference signal.

For the autopilot to hover at an altitude of 10m at a specified x and y location, it is required to add a pitch and roll mechanic that would move the quadcopter to the desired location. The roll motion would move the quadcopter in the y-direction and the pitch motion would move it in the x-direction.

Unknown System Parameter

To better understand the motion of the quadcopter, it is beneficial to use another quadcopter of the same configuration, but with an unknown system parameter. With this setup, it is possible to determine the quadcopter's moment of inertia I_{zz}. To achieve this, autopilot is used, and focus is placed on the yaw. From this, the equation is derived below where $K_d = 1$ and $K_p = 10$.

$$\mathrm{N} = I_{zz}\left(\left(\dot{\psi}_r - \dot{\psi}\right) + (\psi_r - \psi)\right) = K_d\left(\dot{\psi}_r - \dot{\psi}\right) + K_p(\psi_r - \psi)$$

Using Laplace transformation and simplifying both sides then provides the equations below.

$$I_{zz}s^2\psi(s) = K_d s\left(\psi_r(s) - \psi(s)\right) + K_p s\left(\psi_r(s) - \psi(s)\right)$$

$$\frac{\psi(s)}{\psi_r(s)} = \frac{K_d s + K_p}{I_{zz}s^2 + K_d s + K_p}$$

With the transfer function found, it is now possible to determine I_{zz} through examination of the yaw angle response under a step input as seen below in Figure 11.

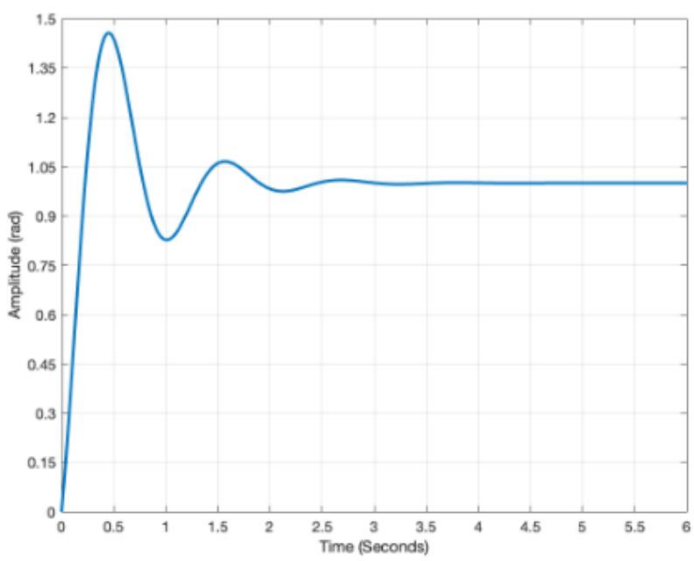

Figure 11. *Quadrotor yaw angle response under a step input.*

As seen in Figure 11, the response provides three vital pieces of information that can be manipulated to solve for I_{zz}. The first of this information is the max amplitude which is where the highest peak is. The second is the steady state amplitude, or the value that the amplitude reaches as time goes towards infinity. The last piece of information is the peak time which is the time at which the max amplitude is reached. The values of these were determined from the graph and are provided below.

$$y_{max} = 1.45$$

$$y_{ss} = 1$$

$$T_p = 0.41$$

Knowing the max amplitude and the steady state amplitude allows the determination of overshoot.

$$\%OS = \frac{y_{max} - y_{ss}}{y_{ss}} 100\% = 0.45$$

With the general form of the transfer function, provided below in equation X, it is possible to determine I_{zz} from our discovered values.

$$\frac{\psi(s)}{\psi_r(s)} = \frac{\omega_n^2}{(s^2 + 2\xi\omega_n s + \omega_n^2)s}$$

To find the values of ξ and ω_n, the following equations are used.

$$T_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$$

$$\%OS = e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}}$$

$$\xi = 0.24634$$

$$\omega_n = 7.9056$$

Finally, the denominators of the transfer functions, general and found, are equated to the found one as seen below in the equation.

$$s^2 + \frac{s}{I_{zz}} + \frac{K_p}{I_{zz}} = s^2 + 2\xi\omega_n s + \omega_n^2$$

This provides the last equation required to solve for I_{zz}.

$$\omega_n^2 = \frac{K_p}{I_{zz}}$$

$$I_{zz} = 0.16\, m^4$$